



Predict, Then Find

Quick facilitation notes & answer key

Goal

Students combine prediction with debugging: predict the intended outcome (both reach 4), run the code, and use the mismatch to locate the bug (C3.2). Bit matches the prediction; Pixel does not (it lands on 3), so Pixel is the buggy script — fix it to 'forward 3'. Predicting first turns debugging into a quick match/no-match check.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: students write the predicted number (4) on a sticky note before running, then compare each sprite's result to it.

Run it (about 20 min)

1. I DO: predict — 'the goal is 4, so both SHOULD land on 4.' Write 4 down before running.
2. WE DO: run Bit (2 forward 2 → 4 — matches) and Pixel (1 forward 2 → 3 — does NOT match). 'Pixel is the bug.'
3. YOU DO: students find Pixel's fix — from 1 it needs forward 3 to reach 4 — and confirm it now matches.
4. Connect: 'Predict first; the script that does not match your prediction is where the bug is.'

Answer key (modelled by the run-gate)

Ex 1 — both SHOULD land on the goal 4 (Bit 4, Pixel 4 if correct).

Ex 2 — Bit: 2 → forward 2 → 4 (matches). Pixel: 1 → forward 2 → 3 (does NOT match). Pixel is the off script.

Ex 3 — Pixel starts at 1 and needs 4, so its move should be forward 3. After the fix, Pixel lands on 4 and matches the goal.

Method note: the prediction (4) is the yardstick; the script whose run differs from it is the buggy one.

Watch for & differentiate

Common error: changing the prediction to match the buggy result — commit the prediction first, in pen.

Common error: suspecting Bit — Bit matched the prediction, so it is correct; the mismatch is in Pixel.

Simplify: write the goal number above each sprite and compare its run to that number.

Extend: ask students to make up a different bug for Pixel that would still miss the goal, then fix it.

Success indicator: the child predicts 4, identifies Pixel as off, and fixes it to forward 3.